June 30, 2026

Dear Editors,

We thank you and the two reviewers for the constructive and detailed evaluation of our manuscript. This submission is a comprehensively revised version of our previous manuscript `symmetry-4339980`, which received a decision of *Major Revision*. We have carefully addressed every comment and substantially strengthened the theoretical justification, the experimental evidence, and the scope of our claims. A point-by-point response to each reviewer is provided in the accompanying response documents, and all changes are marked in the highlighted version of the manuscript. Below we summarize the principal revisions.

Manuscript title: Multi-Granularity Feature Decomposition with Ranking-Aware Regularization for Old Film Restoration

Previous Manuscript ID: symmetry-4339980

Principal Revisions

1. **Main model selection and symmetry justification (Reviewer 2, Comment 1; Reviewer 1, Comments 1 and 3).** Following an internal model-selection review, we froze the main model as **F1 (MGFE+RATR, average fusion)** and demoted the previous max-fusion configuration A3 to a fusion-ablation control. We strengthened the symmetry interpretation by introducing two formal invariance properties (branch permutation and forward/backward direction swap) and by honestly narrowing the claim to functional/representation-level symmetry.

2. **Novelty positioning and difference analysis (Reviewer 1, Comment 1).** We added an architecture $\times$ theory difference table that positions MGFE/RATR against multi-scale pyramids, token-stripe ReID, Mamba-restoration, and diversity-regularization methods along six dimensions, clarifying that the increment over MambaOFR is granularity decomposition plus ranking-aware diversity regularization.

3. **Fair comparison protocol (Reviewer 1, Comment 2).** We adopted a single project-internal retraining protocol under the same pipeline and iteration budget for all compared variants. Enhancement-oriented methods (DeOldify, OldPhoto, DeepRemaster) are explicitly labeled as reference methods rather than task-matched baselines.

4. **Extended ablation study (Reviewer 1, Comment 3; Reviewer 2, Comment 3).** We added single-variable ablations for token insertion (I1), branch aggregation (AG1), fusion strategy (F1 average versus A3 max), ranking metric (R1), and diversity loss (D1). The ablation table now explains the distortion–perception trade-off observed in the MGFE-only configuration and how RATR with average fusion resolves it.

5. **Direct RATR validation (Reviewer 2, Comment 2).** We complemented quantitative results with direct visual evidence: branch activation/response maps and training-time branch-similarity evolution curves comparing the A1 MGFE-only configuration against the F1 full model.

6. **Qualitative figures and supplementary video (Reviewer 1, Comment 4; Reviewer 2, Comments 4 and 8).** We regenerated Figures 4 and 5 with higher-resolution ROIs and three-class annotations (scratch, texture, temporal), and added a supplementary video (Video S1) showing side-by-side restoration comparisons on real-world archival clips.

7. **Efficiency analysis (Reviewer 1, Comment 5).** We quantified parameters, FLOPs, runtime, peak GPU memory, training time, and FPS under a unified protocol for MambaOFR, the proposed F1 model, and transformer baselines (VRT/RVRT/RTN/RRTN).

8. **Failure-case and limitations discussion (Reviewer 1, Comment 6).** We added a focused failure-case analysis comparing F1 against the A0 baseline on the worst-performing clips and expanded the Limitations section.

9. **Reproducibility (Reviewer 2, Comment 5).** We added implementation details on token initialization (std = 0.02), the insertion operator, hyperparameter selection, and the planned public code release.

10. **English language and references (Reviewer 2, Comments 6, 9 and 10).** We shortened repetitive passages in the Introduction, unified terminology, performed a final MDPI-format reference check, and carried out end-to-end English proofreading with a frozen-number gate.

Ethics Statement

All algorithmic experiments were conducted on publicly available video datasets for old-film restoration evaluation. The expert subjective evaluation mentioned in the manuscript was an internal visual-quality scoring of restored frames by internal expert raters; it did not recruit external human subjects, collect human biological samples, record identifiable personal or private information, or involve any intervention on the human body. In accordance with Article 2 of the *Measures for Scientific and Technological Ethics Review (Trial)* (Guo Ke Fa Jian [2023] No. 167), this work does not fall within the scope requiring mandatory ethics review, and no ethics approval or exemption document is required.

Declarations

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with, or published in, another journal. All authors have approved the manuscript and agree with its submission to *Symmetry*. We acknowledge that this work was previously submitted to *Symmetry* (Manuscript ID: symmetry-4339980), and the present resubmission addresses all reviewer and editor comments from that round.

Code, configurations, and evaluation scripts are publicly available at `https://github.com/wscffaa/MgfrOFR`.

Sincerely,

Youdong Ding

(on behalf of all authors)